Variance Reduction for Monte Carlo Pricing of Credit Derivatives

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Contents

1 Introduction

This introduction outlines the project goals, constraints, and contribution. The project is a tutorial on Monte Carlo variance reduction with credit as an application domain. It motivates why stochastic hazards and exotic features push us to simulation, and it previews the toolkit and results.

2 Literature Review

2.1 Scope, Motivation, and Related Literature

Monte Carlo simulation is a central tool in computational finance and applied probability, particularly when analytical valuation becomes infeasible due to stochastic hazards or exotic payoff features. In credit derivatives, closed-form pricing breaks down once the hazard rate becomes stochastic or when coupons and redemptions depend on default outcomes. In such settings, Monte Carlo estimation provides a flexible numerical approach capable of handling discontinuities and complex path dependence. The challenge is not only to obtain unbiased estimators of expected payoffs but to do so efficiently by minimising variance for a fixed computational budget.

The methodological foundation of this project lies in the simulation and variance-reduction literature. Law [6] provides the canonical reference on discrete-event and Monte Carlo simulation methodology, including estimator design, random-number generation, and efficiency-improvement techniques. Glasserman [4] extends these ideas to financial engineering, offering rigorous treatments of convergence diagnostics, variance-reduction algorithms, and pathwise derivative estimation. Together they form the theoretical and algorithmic backbone of this work.

The application context for this project draws on reduced-form credit modelling, where default is represented as a stochastic event governed by a hazard-rate process. Schönbucher [9] presents a practitioner-oriented synthesis of intensity-based credit models and simulation approaches for credit derivatives, while O'Kane and Turnbull [8] outlines the market conventions and bootstrapping techniques used to construct hazard curves and value credit default swaps (CDS). These texts provide the minimal domain knowledge required to translate generic Monte Carlo concepts into a credit-specific test case.

Beyond these four core references, broader treatments of credit risk appear in Duffie and Singleton [3], Lando [5], and Bielecki and Rutkowski [1], each formalising the mathematical foundations of intensity-based and structural models. Brigo and Mercurio [2] complements these works by detailing interest-rate modelling and discounting frameworks relevant to credit-linked products. These additional sources are referenced for completeness but remain outside the primary simulation and variance-reduction focus of this project.

2.2 Pseudo-Random Number Generation

Monte Carlo methods depend on streams of pseudo-random numbers that emulate independent samples from the uniform distribution on $(0, 1)$. Modern simulation packages typically use the Mersenne Twister algorithm [7], a 623-dimensionally equidistributed generator with an extremely long period of $2^{19937} - 1$. NumPy, used in this project, adopts this generator by default. Each simulation uses an explicit seed to guarantee reproducibility, ensuring that all results are replicable. Because the Mersenne Twister already meets the quality criteria recommended by Law [6], no alternative generator is explored. The focus remains on how random draws are exploited within variance-reduction frameworks rather than on the mechanics of the generator itself.

2.3 Monte Carlo Estimation and Error Analysis

The standard Monte Carlo estimator for an expected value $\mu = \mathbb{E}[f(X)]$ is

$$\hat{\mu}_N = \frac{1}{N}\sum_{i=1}^{N} f(X_i), \tag{1}$$

where X_i are independent random draws from the target distribution. By the Law of Large Numbers, $\hat{\mu}_N \to \mu$ almost surely as $N \to \infty$. The Central Limit Theorem implies

$$\sqrt{N}(\hat{\mu}_N - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2), \tag{2}$$

so the standard error declines at rate $1/\sqrt{N}$. This slow convergence motivates the study of variance-reduction techniques that deliver comparable precision with fewer simulated paths [4, 6].

Error analysis decomposes mean-squared error into bias and variance. Since unbiased estimators have zero bias, simulation efficiency is governed mainly by variance. Empirical variance can be estimated via batch means or replication analysis, and confidence intervals quantify estimator precision. Figure **??** may illustrate the typical $1/\sqrt{N}$ convergence pattern.

2.4 Variance Reduction Techniques

Variance-reduction methods exploit structure in the problem to lower estimator variance without introducing bias. Let $\hat{\theta}$ denote a Monte Carlo estimator with variance $\mathrm{Var}[\hat{\theta}]$. A successful technique produces an alternative estimator $\hat{\theta}^*$ satisfying $\mathrm{Var}[\hat{\theta}^*] < \mathrm{Var}[\hat{\theta}]$ for equal computational effort.

Common Random Numbers (CRN). CRN reuses identical random-number streams across related simulations to induce positive correlation. If Y_1 and Y_2 are outputs using the same random seeds,

$$\mathrm{Var}(Y_1 - Y_2) = \mathrm{Var}(Y_1) + \mathrm{Var}(Y_2) - 2\rho\, \sigma_{Y_1}\sigma_{Y_2}.$$

A positive correlation ρ lowers the variance of their difference, improving precision in comparative analyses such as sensitivity tests in credit simulations.

Antithetic Variates. Antithetic sampling generates negatively correlated pairs. For a uniform draw U_i, its complement $1 - U_i$ provides an antithetic pair. When transformed via the Box–Muller or Polar method into standard normals, the resulting $(Z_i, -Z_i)$ pairs yield reduced estimator variance for monotonic payoffs [4]. The method is simple and typically effective.

Stratified and Latin Hypercube Sampling. Stratified sampling divides the uniform $(0, 1)$ range into K equal-probability strata, drawing one sample from each. The estimator becomes

$$\hat{\mu}_{strat} = \frac{1}{K}\sum_{i=1}^{K} f(F^{-1}(U_i)),$$

ensuring coverage across the distribution and reducing clustering in the tails. Latin Hypercube Sampling generalises this idea to multiple dimensions by enforcing stratification along each marginal. The design achieves superior space-filling and typically yields lower variance for smooth response surfaces [6]. Figure **??** can visualise the difference between random sampling and LHS.

Control Variates. Control variates exploit correlation with a secondary variable of known expectation. Given $f(X)$ and $g(X)$ with $\mathbb{E}[g(X)]$ known, the adjusted estimator is

$$\hat{\mu}_{cv} = \bar{f} - b(\bar{g} - \mathbb{E}[g]), \tag{3}$$

where $b^* = \mathrm{Cov}(f,g)/\mathrm{Var}(g)$. Variance reduction improves with correlation strength. In this project, an analytically priced risky bond serves as a control variate for the simulated credit-linked note payoff.

2.5 Application to Credit Derivatives

Credit derivatives provide a structured and intuitive test case for Monte Carlo simulation. A risky bond differs from a risk-free bond by the possibility of default, which halts coupon payments and triggers a recovery amount. A credit default swap transfers this risk: the buyer pays periodic premiums until default or maturity and receives compensation equal to the loss given default. The relationship

$$\text{Risk-Free Bond} = \text{Risky Bond} + \text{CDS}$$

captures this equivalence succinctly [9]. A credit-linked note embeds this exposure in a funded form, offering higher coupons in exchange for assuming the reference entity's default risk. In valuation terms, a CLN is a bond combined with a short CDS position [8]. Deterministic hazard frameworks derive survival from bootstrapped curves, while stochastic frameworks model the hazard as a random process requiring simulation. Variance-reduction methods enhance estimator stability in low-default regimes.

2.6 Implementation and Practical Notes

Simulations are implemented in Python using NumPy. The Mersenne Twister ensures reproducibility, and seeds are fixed for each experiment. Interest-rate and hazard-rate curves are represented by exponential functional forms for tractability. In production these curves would be bootstrapped from traded instruments such as swaps and CDSs, but that process lies outside this project's scope. Calendar conventions and accrual adjustments are abstracted away, since the emphasis is on comparing convergence rates and variance-reduction performance rather than on reproducing full market infrastructure.

2.7 Summary and Literature Gap

While the literatures on Monte Carlo simulation and credit risk are individually mature, few studies apply variance-reduction methods systematically to credit-linked notes or related structures. This project fills that gap by demonstrating how classical techniques can improve estimator precision for stochastic-hazard credit payoffs.

3 Credit Model Setup

3.1 Credit Product Definitions

This section introduces three core instruments used throughout the project: risky bonds, credit default swaps, and credit linked notes. The goal is to give a clear, intuitive explanation that assumes no prior knowledge of credit markets. We first explain how default works, then show how discount factors and survival probabilities enter the valuation of each cashflow. We break each payoff into simple components and give diagrams that illustrate the structure visually.

Default, Recovery, and Survival

A borrower may default at any time before a bond matures. Default means:

- all future coupon payments stop,
- the principal is not paid back in full,
- instead the investor receives a recovery amount R times notional,
- recovery is typically paid immediately when default occurs.

If default happens at time t, this event has probability density $h(t)Q(0,t)$, where $h(t)$ is the hazard rate and $Q(0,t)$ is the survival probability up to t. This explains why recovery terms appear as integrals whereas coupons appear as summations over discrete dates.

Discount factors $P(0,t)$ reduce future cashflows to present value. Survival probabilities $Q(0,t)$ weigh cashflows by the chance that the issuer has not defaulted.

Risky Bonds

A risky bond pays periodic coupons and a principal at maturity, but only if the issuer survives until these dates. If the issuer defaults earlier, coupons stop and the investor receives a recovery payment. We decompose the value of a risky bond into three parts:

1. Coupon leg

$$V_{\text{coupon}} = \sum_i c\, P(0,t_i)\, Q(0,t_i).$$

2. Redemption leg

$$V_{\text{redemption}} = P(0,T)\, Q(0,T).$$

3. Recovery leg

$$V_{\text{recovery}} = \int_0^T R\, P(0,t)\, h(t)\, Q(0,t)\, dt.$$

Total value

$$V_{\text{risky}} = V_{\text{coupon}} + V_{\text{redemption}} + V_{\text{recovery}}.$$

Cashflow diagram

```
Time:   0         t1        t2        ...       T
        |---------|---------|---------|---------|
Cash:     c         c         c         c     +1 (if survive)
Default:            X  ---> Recovery R * Notional
```

Credit Default Swaps

A credit default swap (CDS) transfers the default risk of a reference entity from one party to another. There are two legs.

1. Premium leg

$$V_{\text{prem}} = s \sum_i P(0,t_i)\, Q(0,t_i).$$

2. Protection leg

$$V_{\text{prot}} = (1 - R) \int_0^T P(0,t)\, h(t)\, Q(0,t)\, dt.$$

Fair spread

$$V_{\text{prem}} = V_{\text{prot}}.$$

Intuition

- A protection buyer is short credit risk.
- A protection seller is long credit risk.

Identity Linking Bonds and CDS

$$\text{Risk Free Bond} = \text{Risky Bond} + \text{CDS}.$$

This identity shows how the CDS removes the risky bond's default component.

Credit Linked Notes

A credit linked note embeds a short CDS position into a funded bond:

$$\text{CLN} = \text{Risk Free Bond} - \text{CDS}.$$

The coupon is higher because the investor is selling credit protection. CLNs allow institutions that cannot trade derivatives to take credit risk in a funded format.

3.2 Deterministic Hazard and Short Rate Curves

Default risk is represented through a hazard rate $h(t)$, giving survival

$$Q(0,t) = \exp\left(-\int_0^t h(u)\, du\right).$$

Similarly,

$$P(0,t) = \exp\left(-\int_0^t r(u)\, du\right)$$

gives discounting.

We avoid market calibration and instead specify smooth parametric curves with closed-form integrals.

3.3 Exponential Hazard and Short Rate Curves

$$r(t) = a_r + b_r e^{-c_r t}, \qquad h(t) = a_h + b_h e^{-c_h t}.$$

Integrated forms:

$$Q(0,t) = \exp\Big(-a_r t - \frac{b_r}{c_r}(1 - e^{-c_r t})\Big),$$

$$S(0,t) = \exp\Big(-a_h t - \frac{b_h}{c_h}(1 - e^{-c_h t})\Big).$$

These deterministic curves will become the *initial conditions* for the HJM model.

3.4 Interpretation of the Parameters

Long-run level a sets the asymptotic intensity/rate.

Shift b controls slope at $t = 0$.

Speed c determines exponential decay, equivalently the half-life:

$$t_{1/2} = \frac{\ln 2}{c}.$$

3.5 Chosen Parameters and Economic Motivation

The project uses simple but realistic parameter sets for the short rate and for two credit qualities: an investment grade (IG) name and a high yield (HY) name. These curves are chosen for plausibility rather than calibration but retain appropriate risk levels and term structure slopes.

Short rate parameters

$$a_r = 0.02, \qquad b_r = 0.02, \qquad c_r = 0.35.$$

This yields $r(0) = 4\%$ and a long run level of 2%. The half life is approximately two years. This generates a mild downward slope that is consistent with many developed market environments.

Investment grade hazard rate

$$a_h = 0.015, \qquad b_h = -0.012, \qquad c_h = 0.30.$$

This gives $h(0) = 0.003$ and a long run intensity of 0.015. The curve slopes upward with a half life near 2.3 years. This matches the pattern of IG credit risk where short horizon default risk is very small but long run uncertainty is higher.

High yield hazard rate

$$a_h = 0.03, \qquad b_h = -0.02, \qquad c_h = 0.40.$$

This yields $h(0) = 0.01$ and a long run intensity of 0.03. The slope is steeper and the half life is shorter (about 1.7 years). This reflects the behaviour of BB or B rated credits with materially higher default risk.

3.6 Spread Interpretation Under Recovery

Under reduced form modelling with recovery of par, the credit spread associated with a hazard rate is

$$s(t) = (1 - R)\, h(t), \tag{4}$$

and we adopt $R = 40\%$ which is standard for unsecured corporates.

Investment grade spreads

$$s_{\mathrm{IG}}(0) = 0.6 \times 0.003 = 18 \text{ bps},$$
$$s_{\mathrm{IG}}(\infty) = 0.6 \times 0.015 = 90 \text{ bps}.$$

High yield spreads

$$s_{\mathrm{HY}}(0) = 0.6 \times 0.01 = 60 \text{ bps},$$
$$s_{\mathrm{HY}}(\infty) = 0.6 \times 0.03 = 180 \text{ bps}.$$

These spread levels are reasonable for IG and HY credit and support the qualitative interpretation of the chosen parameters.

3.7 HJM Framework for Rates and Hazards

The deterministic short rate curve $r(t)$ and hazard rate curve $h(t)$ provide the initial term structures of discount and survival. The Heath–Jarrow–Morton (HJM) approach takes these initial curves as inputs and specifies arbitrage-free stochastic dynamics for the entire forward curve. In HJM models the forward curve is the fundamental state variable and all randomness enters through its evolution rather than through the initial term structure.

Short Rate, Forward Rate, and Their Relationships

The short rate $r(t)$ represents the instantaneous risk-neutral interest rate at time t. The price of a zero-coupon bond maturing at T is

$$P(t,T) = \mathbb{E}_t\left[\exp\Big(-\int_t^T r_u\,du\Big)\right].$$

The instantaneous forward rate $f(t,T)$ is defined by

$$f(t,T) = -\frac{\partial}{\partial T}\log P(t,T),$$

which represents the rate applied over an infinitesimal period at time T. The short rate is recovered from the diagonal:

$$r_t = f(t,t).$$

When $r(t)$ is deterministic,

$$P(0,T) = \exp\Big(-\int_0^T r(u)\,du\Big), \qquad f(0,T) = r(T).$$

Thus the deterministic short rate curve is equal to the initial forward curve.

Hazard Rate, Forward Hazard, and Their Relationships

The hazard rate $h(t)$ is the instantaneous default intensity. Survival to T is

$$Q(t,T) = \mathbb{E}_t\left[\exp\Big(-\int_t^T h_u\,du\Big)\right].$$

The forward hazard rate is

$$\gamma(t,T) = -\frac{\partial}{\partial T}\log Q(t,T), \qquad h_t = \gamma(t,t).$$

In the deterministic case,

$$Q(0,T) = \exp\Big(-\int_0^T h(u)\,du\Big), \qquad \gamma(0,T) = h(T).$$

The deterministic hazard curve is therefore the initial forward hazard surface.

Initial Forward Surfaces for HJM

The exponential deterministic curves derived earlier give

$$f(0,T) = r(T), \qquad \gamma(0,T) = h(T).$$

These surfaces serve as the initial conditions for the HJM evolution.

HJM Dynamics and Drift Conditions

HJM specifies forward dynamics of the form

$$df(t,T) = \alpha_r(t,T)\,dt + \sigma_r(t,T)\,dW_t^{(r)},$$

$$d\gamma(t,T) = \alpha_h(t,T)\,dt + \sigma_h(t,T)\,dW_t^{(h)},$$

with instantaneous correlation

$$dW_t^{(r)}dW_t^{(h)} = \rho\,dt.$$

We adopt one-factor exponential volatilities,

$$\sigma_r(t,T) = \sigma_r e^{-c_r(T-t)}, \qquad \sigma_h(t,T) = \sigma_h e^{-c_h(T-t)}.$$

The HJM drift restriction ensures absence of arbitrage. It gives

$$\alpha_r(t,T) = \sigma_r(t,T)\int_t^T \sigma_r(t,s)\,ds = \frac{\sigma_r^2}{2c_r}\big(1 - e^{-2c_r(T-t)}\big),$$

$$\alpha_h(t,T) = \sigma_h(t,T)\int_t^T \sigma_h(t,s)\,ds = \frac{\sigma_h^2}{2c_h}\big(1 - e^{-2c_h(T-t)}\big).$$

Forward surfaces evolve as

$$f(t,T) = f(0,T) + \int_0^t \alpha_r(u,T)\,du + \int_0^t \sigma_r(u,T)\,dW_u^{(r)},$$

$$\gamma(t,T) = \gamma(0,T) + \int_0^t \alpha_h(u,T)\,du + \int_0^t \sigma_h(u,T)\,dW_u^{(h)}.$$

Interpretation of Forward Curve Evolution

The HJM framework models the time evolution of the entire forward curve. The forward rate $f(t,T)$ is defined for maturities $T \geq t$, and at each observation time t the function $T \mapsto f(t,T)$ describes the market-implied instantaneous rates for all remaining maturities. As time increases, both the domain and the shape of this forward curve change.

At time 0 the full forward curve

$$f(0,T), \qquad T \in [0,T^*],$$

is observable from the initial term structure. This curve represents the market-implied future short rates from the present time.

At a later time t_1 one observes a new forward curve

$$f(t_1,T), \qquad T \geq t_1.$$

The portion of the original curve with maturities $T < t_1$ is no longer relevant because those maturities lie in the past. The short rate at time t_1 is obtained from the diagonal through

$$r(t_1) = f(t_1,t_1).$$

At a later time t_2 the entire curve evolves again,

$$f(t_2,T), \qquad T \geq t_2,$$

and both its location and its shape change due to the stochastic HJM dynamics. The curve now begins at the point $T = t_2$, and its behaviour for $T > t_2$ reflects the accumulated randomness up to time t_2.

This picture highlights several key properties that are central to HJM modelling.

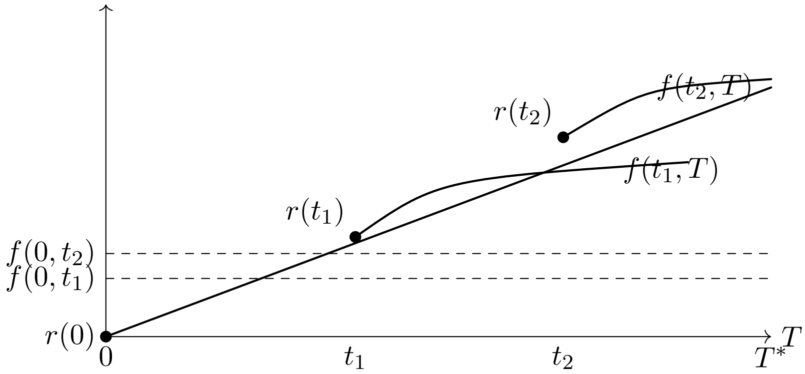

Figure 1: Evolution of the forward curve. At time zero, the forward curve $f(0, T)$ is defined for maturities in $[0, T^*]$ and the short rate is $r(0) = f(0, 0)$. At time t_1 the forward curve $f(t_1, T)$ is defined for maturities in $[t_1, T^*]$, and the short rate is $r(t_1) = f(t_1, t_1)$.

- The forward curve evolves as a random function. At times t_0, t_1, and t_2 the entire mapping $T \mapsto f(t, T)$ shifts according to the dynamics implied by the forward SDE.
- The short rate always lies on the diagonal $T = t$ and satisfies $r(t) = f(t, t)$. This identity is fundamental because the short rate drives discounting and appears in all risk-neutral valuation formulas.
- As time increases, the region $T < t$ becomes irrelevant. The forward curve is only defined for maturities that remain in the future relative to the current time.
- HJM models the full forward curve rather than the short rate. The short rate is recovered from the diagonal, but the dynamics are specified for the entire surface $f(t, T)$.
- The simulated object in a Monte Carlo implementation is therefore a curve-valued stochastic process. For every simulation path and every time step, one obtains a full forward curve $f(t, T)$ for all $T \geq t$.

These ideas extend directly to the forward hazard surface $\gamma(t, T)$ used for credit modelling. In the CLN valuation framework the HJM dynamics generate stochastic forward hazard curves. At each time step the instantaneous hazard rate is extracted from the diagonal through

$$h_t = \gamma(t, t),$$

and this quantity drives CDS spreads, survival probabilities, barrier conditions, optional redemptions, and any payoff components that depend on credit quality. Forward hazard evolution ensures that the expected survival curve matches the deterministic initial curve, which is essential for unbiased Monte Carlo pricing.

Understanding the evolution of the forward surfaces $f(t, T)$ and $\gamma(t, T)$ clarifies the structure of the simulation. It explains why diagonals determine the instantaneous rates, why pathwise integrals of these rates appear in discount and survival factors, and how deterministic term structures propagate consistently into stochastic valuation. This conceptual view forms the link between deterministic credit curves, stochastic forward hazard modelling, and the pathwise hazard rates used in credit-linked note pricing.

Bond Prices, Survival, and Instantaneous Rates

The HJM evolution provides the entire forward surfaces $f(t,T)$ and $\gamma(t,T)$ for interest rates and hazard rates. From these surfaces we recover bond prices, survival probabilities, and the instantaneous short rate and spot hazard rate. These quantities are needed for Monte Carlo valuation of credit-linked notes.

Recovering bond prices and survival probabilities. Given forward surfaces at time t, the pathwise zero-coupon bond price and survival value are

$$P(t,T) = \exp\Big(-\int_t^T f(t,u)\,du \Big), \qquad Q(t,T) = \exp\Big(-\int_t^T \gamma(t,u)\,du \Big).$$

These identities hold in both deterministic and stochastic settings. The HJM drift restriction ensures

$$\mathbb{E}\left[P(t,T)\right] = P(0,T), \qquad \mathbb{E}\left[Q(t,T)\right] = Q(0,T),$$

which guarantees that the simulated curves reproduce the initial term structures in expectation.

Instantaneous rates along the path The short rate and spot hazard rate are obtained from the diagonals:

$$r_t = f(t,t), \qquad h_t = \gamma(t,t).$$

Since negative hazard rates are not meaningful, we apply

$$h_t \leftarrow \max\{h_t, 0\}.$$

The accumulated discount factor and survival factor along a simulation path are

$$D(0,t) = \exp\Big(-\int_0^t r_u\,du \Big), \qquad S(0,t) = \exp\Big(-\int_0^t h_u\,du \Big).$$

Why the pathwise integrals are required. Many exotic credit products depend on the instantaneous hazard rate or the cumulative hazard along the path. For example, model-implied CDS spreads satisfy

$$s_t = (1-R)\,h_t,$$

so any payoff referencing credit spreads must use the simulated h_t at each step. Callable and barrier CLNs may depend on the survival value $Q(t,T)$ or on the mark-to-market spread at call dates. Range accrual CLNs often require the hazard rate to remain inside or outside a region.

Even when a payoff depends only on credit quantities, the risk-neutral valuation formula requires discounting through

$$D(0,t) = \exp\Big(-\int_0^t r_u\,du \Big),$$

so the short rate must also be accumulated along the path.

Role of the forward SDEs in simulation. The forward dynamics

$$df(t,T) = \alpha_r(t,T)\,dt + \sigma_r(t,T)\,dW_t^{(r)}, \qquad d\gamma(t,T) = \alpha_h(t,T)\,dt + \sigma_h(t,T)\,dW_t^{(h)},$$

are discretised in time in the Monte Carlo simulation. At each step the entire forward curve is updated, and the diagonal values $r_t = f(t,t)$ and $h_t = \gamma(t,t)$ are extracted. Numerical integration of these processes yields $D(0,t)$ and $S(0,t)$, which drive the pricing of all subsequent CLN cashflows.

The implementation details, including discretisation schemes and variance reduction, are presented in the following section.

4 Monte Carlo Simulation Framework

Monte Carlo simulation is required whenever payoffs depend on entire paths of the short rate and hazard rate. Credit linked notes often contain features such as range accrual coupons, knockouts on the redemption payment, or digital adjustments to principal. These features depend on the evolution of (r_t, h_t) across time rather than on their terminal values only. Analytic integration is not possible once the payoff depends on the trajectory. The HJM framework introduced earlier provides a consistent way to generate stochastic paths for the forward surface. From these paths we extract (r_t, h_t) along the diagonal and compute discount factors and survival probabilities path by path.

The structure of this section is as follows. Section 4.1 describes the Monte Carlo estimator and sampling method. Section 4.2 describes error diagnostics. Section 4.3 introduces variance reduction techniques. Section 4.4 develops the control variate methodology designed specifically for CLNs.

4.1 Why Monte Carlo is Needed for CLNs

Simple risky bonds and vanilla CLNs with constant coupons are not path dependent. Their values can be computed analytically using discount factors and survival probabilities. Exotic CLN features break this structure. Examples include:

- **Range accrual coupons**. The coupon at t_i depends on the fraction of dates in $[t_{i-1}, t_i]$ for which $\gamma(t, t) \in [L, U]$.
- **Knockout redemption**. The principal is paid only if the hazard stays below a barrier over the life of the note.
- **Digital jumps**. The redemption amount depends on whether the hazard rate crosses a threshold before a specific date.
- **Stochastic discounting and survival**. Under the full HJM model, the discount factor and survival probability are path integrals of stochastic processes rather than deterministic functions.

These features make the payoff depend on infinitely many points of the path. Monte Carlo methods are therefore unavoidable.

4.2 Sampling and Estimator Construction

Let Y be the discounted payoff of the CLN under the risk neutral measure. Monte Carlo simulation generates N independent replicas $Y^{(1)}, \dots, Y^{(N)}$. The basic estimator is

$$\hat{Y}_N = \frac{1}{N} \sum_{i=1}^{N} Y^{(i)}.$$

Central limit theorem Under mild regularity conditions:

$$\sqrt{N}\big(\hat{Y}_N - \mathbb{E}[Y]\big) \to \mathcal{N}\big(0, \mathrm{Var}(Y)\big).$$

Variance of the estimator

$$\mathrm{Var}(\hat{Y}_N) = \frac{1}{N}\,\mathrm{Var}(Y).$$

Generation of correlated normals

The HJM dynamics involve two Brownian motions with correlation ρ. To generate correlated shocks:

$$Z_r = U, \qquad Z_h = \rho U + \sqrt{1-\rho^2}\, V,$$

where U and V are independent standard normals. At each time step, these shocks are inserted into the Euler or Milstein discretisation schemes described below.

4.3 Discretisation of the HJM Dynamics

The HJM framework produces forward rate processes $f(t,T)$ and $\gamma(t,T)$ for $T > t$. To simulate short rate and hazard paths we evolve these forward surfaces across a rectangular time-maturity grid.

Time discretisation

Let the simulation horizon be $[0, T_{\max}]$ with a time grid:

$$0 = t_0 < t_1 < \cdots < t_M = T_{\max}, \qquad \Delta t = t_{k+1} - t_k.$$

For each t_k we store the values of the forward curve at maturities $T_j \geq t_k$. The diagonal elements give:

$$r_{t_k} = f(t_k, t_k), \qquad h_{t_k} = \max\{\gamma(t_k, t_k), 0\}.$$

Euler discretisation for forward rates

For each maturity $T_j > t_k$:

$$f(t_{k+1}, T_j) = f(t_k, T_j) + \alpha_r(t_k, T_j)\,\Delta t + \sigma_r(t_k, T_j)\sqrt{\Delta t}\, Z_r.$$

A similar expression holds for the forward hazard curve:

$$\gamma(t_{k+1}, T_j) = \gamma(t_k, T_j) + \alpha_h(t_k, T_j)\,\Delta t + \sigma_h(t_k, T_j)\sqrt{\Delta t}\, Z_h.$$

Euler is stable for forward rates because their diffusion is normal.

Milstein scheme for the hazard diagonal

Forward hazards follow a normal diffusion. The spot hazard $h_t = \gamma(t,t)$ however represents the instantaneous intensity. To reduce downward drift and keep the hazard positive we apply a simple Milstein correction:

$$h_{t_{k+1}} = h_{t_k} + \alpha_h(t_k, t_k)\,\Delta t + \sigma_h(t_k, t_k)\sqrt{\Delta t}\, Z_h + \frac{1}{2}\sigma_h(t_k, t_k)\frac{\partial \sigma_h}{\partial h}(t_k, t_k)\Delta t(Z_h^2 - 1).$$

Since $\sigma_h(t,T) = \sigma_h e^{-c_h(T-t)}$ does not depend on h, the derivative term is zero. For completeness we include a positivity clamp:

$$h_{t_{k+1}} \leftarrow \max\{h_{t_{k+1}}, 0\}.$$

Numerical construction of discount factors and survival probabilities

Along each simulated path:

$$D(0, t_{k+1}) = D(0, t_k) \exp\big(- r_{t_k} \Delta t \big),$$

$$S(0, t_{k+1}) = S(0, t_k) \exp\big(- h_{t_k} \Delta t \big).$$

These pathwise discount and survival measures feed directly into the CLN payoff calculator.

4.4 Error Analysis and Diagnostics

The sample variance estimator is

$$\hat{\sigma}_N^2 = \frac{1}{N-1} \sum_{i=1}^{N} \big(Y^{(i)} - \hat{Y}_N \big)^2.$$

A $(1 - \alpha)$ confidence interval is

$$\hat{Y}_N \pm z_{1-\alpha/2} \frac{\hat{\sigma}_N}{\sqrt{N}}.$$

Diagnostic techniques

- **Running mean plot**. Reveals slow convergence or drift.
- **Batch means**. Break one large sample into K batches.
- **Independent replications**. Repeat the entire simulation with different seeds.

These diagnostics help detect correlation, instability or variance explosion under extreme parameters.

4.5 Variance Reduction Techniques

Variance reduction is essential for CLN pricing because exotic coupons may have high variance when simulated directly.

Common random numbers

For two related payoffs X and Y using the same random numbers:

$$\mathrm{Var}(X - Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\,\mathrm{Cov}(X, Y).$$

CLNs with exotic coupons are highly correlated with vanilla CLNs under HJM. CRN therefore produce substantial variance reduction when estimating differences.

Antithetic variates

Given shocks Z, also simulate with $-Z$:

$$\hat{Y}^{\mathrm{anti}} = \frac{1}{2} \big(Y(Z) + Y(-Z) \big).$$

This reduces variance when the payoff is monotone in the shocks. For credit payoffs monotonicity often holds approximately.

Stratified sampling and Latin hypercube sampling

Stratified sampling partitions $[0, 1]$ into K intervals with one sample drawn from each. Mapping these to normal variates ensures better space filling.

Latin hypercube sampling extends stratification to multiple dimensions. It is particularly beneficial when simulating high dimensional hazard and rate paths under HJM.

Control variates

Given a control C with known expectation:

$$\hat{Y}_N^{\text{cv}} = \hat{Y}_N + b(\hat{C}_N - \mathbb{E}[C]),$$

with optimal

$$b^\star = -\frac{\operatorname{Cov}(Y, C)}{\operatorname{Var}(C)}.$$

4.6 Control Variate Design for CLNs

Under the HJM framework, the expected Monte Carlo price of a vanilla CLN equals the analytic price obtained from the deterministic curves. This property is essential. It means the vanilla CLN is an ideal control variate for any exotic CLN with the same maturity and reference name.

Let Y be the exotic CLN payoff and let C be the vanilla CLN payoff on the same path. The analytic vanilla price is denoted V_{van}.

$$\hat{Y}_N^{\text{cv}} = \hat{Y}_N + b(\hat{C}_N - V_{\text{van}}).$$

Practical estimation of b Estimate b using sample covariance:

$$b = -\frac{\widehat{\operatorname{Cov}}(Y, C)}{\widehat{\operatorname{Var}}(C)}.$$

Why this control works extremely well

- The exotic CLN differs from the vanilla CLN in only a small region of state space where exotic triggers activate.
- When the hazard and rate volatilities are modest, most simulated paths behave like a vanilla CLN.
- Correlation between Y and C is very high, often above 0.95.
- HJM ensures $\mathbb{E}[C] = V_{\text{van}}$, so the control variate is perfectly aligned in expectation.

The resulting variance reduction is dramatic, typically reducing error by one or two orders of magnitude.

5 Numerical Experiments and Results

Present quantitative comparisons, variance reduction factors, and convergence plots for the techniques across toy problems and credit payoffs.

6 Discussion of Results

Interpret the results. Explain when each method performs well or poorly. Discuss correlation strength for control variates and monotonicity for antithetic.

7 Exotic Extensions

Outline conceptual extensions such as range accrual features or multi-factor hazard models. Keep scope conceptual.

8 Implementation Notes

8.1 Environment and Dependencies

List Python and NumPy versions, Matplotlib, and any other packages. State the fixed RNG seed.

8.2 Reproducibility Recipe

Give a step-by-step recipe to reproduce results: set seed, generate uniforms, transform variates, run estimators, record CIs, and plot funnels. Avoid long code listings.

9 Conclusion

Summarise learning outcomes and empirical findings. Mention potential future work such as importance sampling or quasi-Monte Carlo if desired.

10 Appendices

Optional stub for any extended derivations or data tables if you choose to include them later.

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